

Week 7: Student Performance – Data Cleaning & Transformation

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Aim: Clean, transform and prepare the student performance dataset in Power BI.

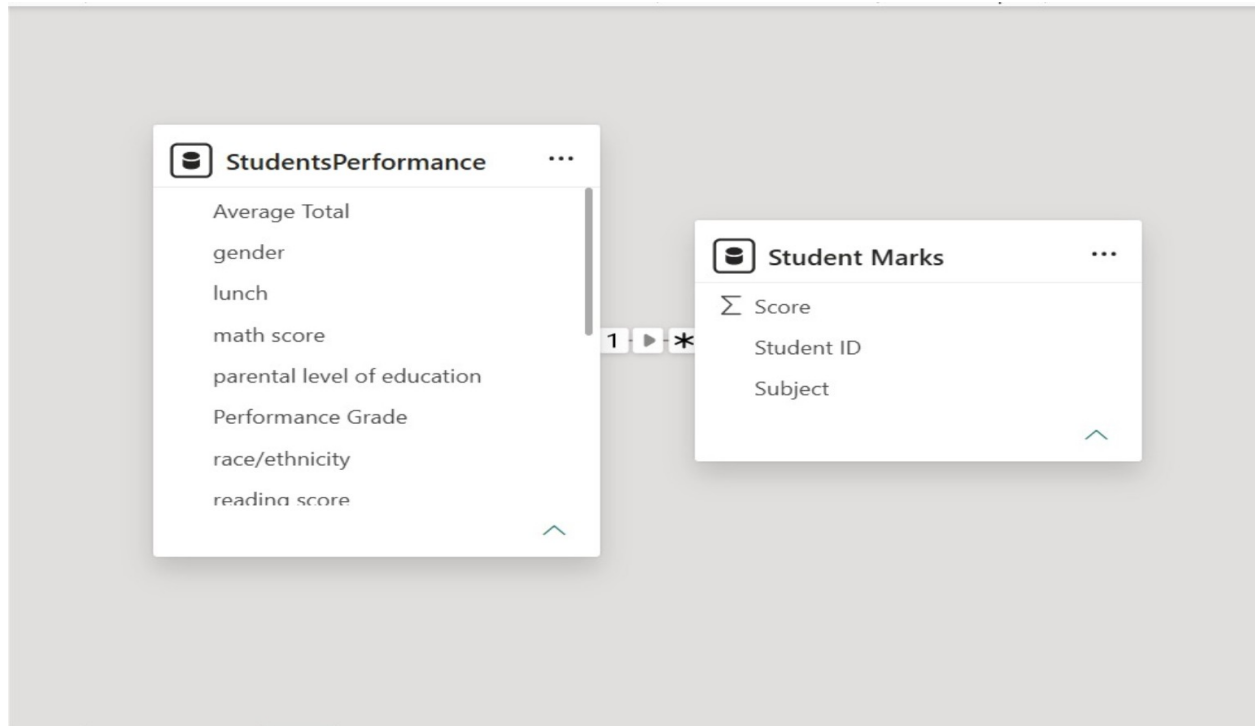
Procedure Summary:

Imported dataset, inspected data types, cleaned records, standardized categorical values, created Total Marks, Average Total, Performance Grade, Result, unpivoted subject columns, created Student Marks table, established relationship, loaded model and created DAX measures.

Relationship between StudentsPerformance and Student Marks.

	AB gender	AB race/ethnicity	AB parental level of education	AB lunch	AB test preparation course
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	female	group B	Bachelor'S Degree	standard	none
2	female	group C	Some College	standard	completed
3	female	group B	Master'S Degree	standard	none
4	male	group A	Associate'S Degree	free/reduced	none
5	male	group C	Some College	standard	none
6	female	group B	Associate'S Degree	standard	none
7	female	group B	Some College	standard	completed
8	male	group B	Some College	free/reduced	none
9	male	group D	High School	free/reduced	completed
10	female	group B	High School	free/reduced	none
11	male	group C	Associate'S Degree	standard	none
12	male	group D	Associate'S Degree	standard	none
13	female	group B	High School	standard	none
14	male	group A	Some College	standard	completed
15	female	group A	Master'S Degree	standard	none
16	female	group C	Some High School	standard	none
17	male	group C	High School	standard	none
18	female	group B	Some High School	free/reduced	none
19	male	group C	Master'S Degree	free/reduced	completed
20	female	group C	Associate'S Degree	free/reduced	none
21	male	group D	High School	standard	none
22	female	group B	Some College	free/reduced	completed

Original dataset in Power Query.



Calculated columns including Total Marks, Average Total, Grade, Result and Student ID.

	ABC 123	Total Marks	1.2	Average Total	ABC 123	Performance Grade	ABC 123	Result	1.2	Student ID
		100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%	Valid Error Empty	100% 0% 0%
1		74		218		72.66666667	A	Pass		1
2		88		247		82.33333333	B+	Pass		2
3		93		278		92.66666667	A+	Pass		3
4		44		148		49.33333333	B	Pass		4
5		75		229		76.33333333	A	Pass		5
6		78		232		77.33333333	A	Pass		6
7		92		275		91.66666667	A+	Pass		7
8		39		122		40.66666667	B	Pass		8
9		67		195		65	A	Pass		9
10		50		148		49.33333333	B	Pass		10
11		52		164		54.66666667	B	Pass		11
12		43		135		45	B	Pass		12
13		73		219		73	A	Pass		13
14		70		220		73.33333333	A	Pass		14
15		58		161		53.66666667	B	Pass		15
16		78		222		74	A	Pass		16
17		86		263		87.66666667	B+	Pass		17
18		28		78		26	F	Fail		18
19		46		134		44.66666667	B	Pass		19
20		61		173		57.66666667	B	Pass		20
21		63		198		66	A	Pass		21
22		70		210		70	A	Pass		22
23										

Verified transformed columns.

	1 ² ₃ writing score	ABC 123 Total Marks	1.2 Average Total	ABC 123 Performance Grade	ABC 123 Result
	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty
1	72	74	218	72.66666667	A
2	90	88	247	82.33333333	B+
3	95	93	278	92.66666667	A+
4	57	44	148	49.33333333	B
5	78	75	229	76.33333333	A
6	83	78	232	77.33333333	A
7	95	92	275	91.66666667	A+
8	43	39	122	40.66666667	B
9	64	67	195	65	A
10	60	50	148	49.33333333	B
11	54	52	164	54.66666667	B
12	52	43	135	45	B
13	81	73	219	73	A
14	72	70	220	73.33333333	A
15	53	58	161	53.66666667	B
16	75	78	222	74	A
17	89	86	263	87.66666667	B+
18	32	28	78	26	F
19	42	46	134	44.66666667	B
20	58	61	173	57.66666667	B
21	69	63	198	66	A
22	75	70	210	70	A
23					

Average Total and Grade verification.

	1 ² ₃ reading score	1 ² ₃ writing score	ABC 123 Total Marks	1.2 Average Total	ABC 123 Performance Grade
	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty	100% 0% 0% ● Valid ● Error ● Empty
1	72	72	74	218	72.66666667
2	69	90	88	247	82.33333333
3	90	95	93	278	92.66666667
4	47	57	44	148	49.33333333
5	76	78	75	229	76.33333333
6	71	83	78	232	77.33333333
7	88	95	92	275	91.66666667
8	40	43	39	122	40.66666667
9	64	64	67	195	65
10	38	60	50	148	49.33333333
11	58	54	52	164	54.66666667
12	40	52	43	135	45
13	65	81	73	219	73
14	78	72	70	220	73.33333333
15	50	53	58	161	53.66666667
16	69	75	78	222	74
17	88	89	86	263	87.66666667
18	18	32	28	78	26
19	46	42	46	134	44.66666667
20	54	58	61	173	57.66666667
21	66	69	63	198	66
22	65	75	70	210	70
23					

Average Total numeric type.

= Table.TransformColumnTypes(#"Added Custom1",{{"Average Total", type number}})									
course	123 math score	123 reading score	123 writing score	ABC 123 Total Marks	1.2 Average Total				
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty				
1		72	72	74	218				
2		69	90	88	247				
3		90	95	93	278				
4		47	57	44	148				
5		76	78	75	229				
6		71	83	78	232				
7		88	95	92	275				
8		40	43	39	122				
9		64	64	67	195				
10		38	60	50	148				
11		58	54	52	164				
12		40	52	43	135				
13		65	81	73	219				
14		78	72	70	220				
15		50	53	58	161				
16		69	75	78	222				
17		88	89	86	263				
18		18	32	28	78				
19		46	42	46	134				
20		54	58	61	173				
21		66	69	63	198				
22		65	75	70	210				
23									

Total Marks calculation.

test preparation course	123 math score	123 reading score	123 writing score	ABC 123 Total Marks
100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1	none	72	72	218
2	completed	69	90	247
3	none	90	95	278
4	none	47	57	148
5	none	76	78	229
6	none	71	83	232
7	completed	88	95	275
8	none	40	43	122
9	completed	64	64	195
10	none	38	60	148
11	none	58	54	164
12	none	40	52	135
13	none	65	81	219
14	completed	78	72	220
15	none	50	53	161
16	none	69	75	222
17	none	88	89	263
18	none	18	32	78
19	completed	46	42	134
20	none	54	58	173
21	none	66	69	198
22	completed	65	75	210
23				

DAX Measures:

Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score and Overall Pass %. These measures enable subject-wise and overall analysis.

Result: The dataset was successfully cleaned, transformed and prepared for visualization in Power BI. Relationships, calculated columns and DAX measures were created successfully.